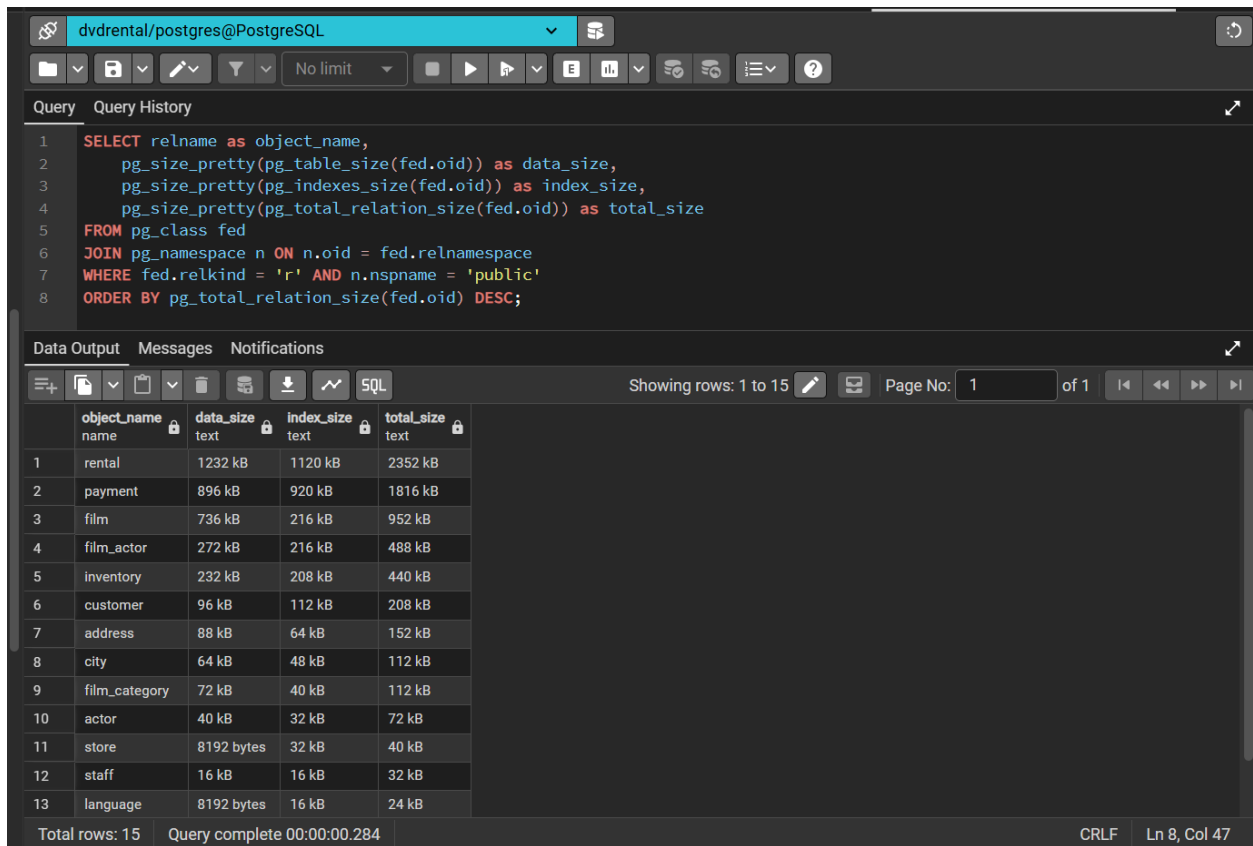


## Lab 1



The screenshot shows a PostgreSQL query editor with the following SQL query:

```
1 SELECT relname as object_name,
2     pg_size_pretty(pg_table_size(fed.oid)) as data_size,
3     pg_size_pretty(pg_indexes_size(fed.oid)) as index_size,
4     pg_size_pretty(pg_total_relation_size(fed.oid)) as total_size
5 FROM pg_class fed
6 JOIN pg_namespace n ON n.oid = fed.relnamespace
7 WHERE fed.relkind = 'r' AND n.nspname = 'public'
8 ORDER BY pg_total_relation_size(fed.oid) DESC;
```

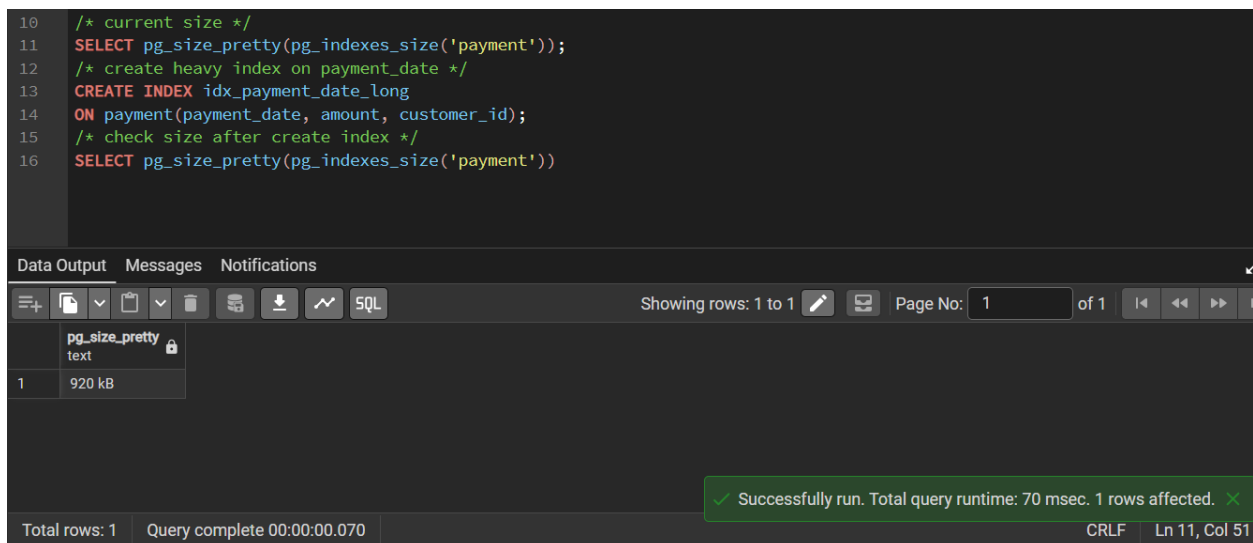
The query results are displayed in a table with the following columns: object\_name, data\_size, index\_size, and total\_size. The results are sorted by total\_size in descending order.

|    | object_name   | data_size  | index_size | total_size |
|----|---------------|------------|------------|------------|
| 1  | rental        | 1232 kB    | 1120 kB    | 2352 kB    |
| 2  | payment       | 896 kB     | 920 kB     | 1816 kB    |
| 3  | film          | 736 kB     | 216 kB     | 952 kB     |
| 4  | film_actor    | 272 kB     | 216 kB     | 488 kB     |
| 5  | inventory     | 232 kB     | 208 kB     | 440 kB     |
| 6  | customer      | 96 kB      | 112 kB     | 208 kB     |
| 7  | address       | 88 kB      | 64 kB      | 152 kB     |
| 8  | city          | 64 kB      | 48 kB      | 112 kB     |
| 9  | film_category | 72 kB      | 40 kB      | 112 kB     |
| 10 | actor         | 40 kB      | 32 kB      | 72 kB      |
| 11 | store         | 8192 bytes | 32 kB      | 40 kB      |
| 12 | staff         | 16 kB      | 16 kB      | 32 kB      |
| 13 | language      | 8192 bytes | 16 kB      | 24 kB      |

Total rows: 15 | Query complete 00:00:00.284 | CRLF | Ln 8, Col 47

## Lab 2

/\* current size \*/



The screenshot shows a PostgreSQL query editor with the following SQL query:

```
10 /* current size */
11 SELECT pg_size_pretty(pg_indexes_size('payment'));
12 /* create heavy index on payment_date */
13 CREATE INDEX idx_payment_date_long
14 ON payment(payment_date, amount, customer_id);
15 /* check size after create index */
16 SELECT pg_size_pretty(pg_indexes_size('payment'))
```

The query results are displayed in a table with the following columns: pg\_size\_pretty. The results show the current size of the payment table.

|   | pg_size_pretty |
|---|----------------|
| 1 | 920 kB         |

Total rows: 1 | Query complete 00:00:00.070 | CRLF | Ln 11, Col 51

Successfully run. Total query runtime: 70 msec. 1 rows affected.

/\* create heavy index on payment\_date \*/

```
10 /* current size */
11 SELECT pg_size_pretty(pg_indexes_size('payment'));
12 /* create heavy index on payment_date */
13 CREATE INDEX idx_payment_date_long
14 ON payment(payment_date, amount, customer_id);
15 /* check size after create index */
16 SELECT pg_size_pretty(pg_indexes_size('payment'))
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 60 msec.

✓ Query returned successfully in 60 msec. ✕

Total rows: Query complete 00:00:00.060 CRLF Ln 14, Col 47

/\* check size after create index \*/

```
10 /* current size */
11 SELECT pg_size_pretty(pg_indexes_size('payment'));
12 /* create heavy index on payment_date */
13 CREATE INDEX idx_payment_date_long
14 ON payment(payment_date, amount, customer_id);
15 /* check size after create index */
16 SELECT pg_size_pretty(pg_indexes_size('payment'))
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

|   | pg_size_pretty<br>text |
|---|------------------------|
| 1 | 1488 kB                |

✓ Successfully run. Total query runtime: 42 msec. 1 rows affected. ✕

Total rows: 1 Query complete 00:00:00.042 CRLF Ln 16, Col 51

Discussion: New index consumes  $(1488 - 920)/8 = 71$  pages

### Lab 3

/\* Create partial index \*/

```
20 /* Create partial index */
21 CREATE INDEX idx_expensive_films
22 ON film(replacement_cost)
23 WHERE replacement_cost > 25.00;
24
25 /* Find size of this specific index */
26 SELECT relname, pg_size_pretty(pg_relation_size(relid))
27 FROM pg_stat_user_indexes
28 WHERE relname = 'idx_expensive_films';
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 57 msec.

Total rows: Query complete 00:00:00.057 CRLF Ln 23, Col 32

/\* Find size of this specific index \*/

```
20 /* Create partial index */
21 CREATE INDEX idx_expensive_films
22 ON film(replacement_cost)
23 WHERE replacement_cost > 25.00;
24
25 /* Find size of this specific index */
26 SELECT relname, pg_size_pretty(pg_relation_size(oid))
27 FROM pg_class
28 WHERE relname = 'idx_expensive_films';
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

|   | relname<br>name      | pg_size_pretty<br>text |
|---|----------------------|------------------------|
| 1 | idx_expensive_fil... | 16 kB                  |

✓ Successfully run. Total query runtime: 62 msec. 1 rows affected. ✕

Total rows: 1 Query complete 00:00:00.062 CRLF Ln 28, Col 39

Task: 16kB of this specific index is very small compare to 568kB of hypothetical full index on replacement\_cost.